INTERVIEW CHEAT SHEET

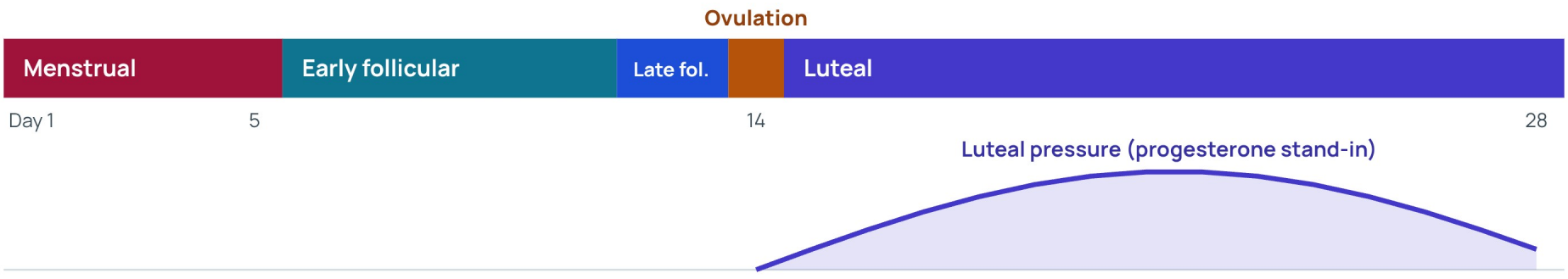
Cycle-Synced Nutrition Engine

Genotype + estimated cycle phase gives an exploratory carb target (PCOS)

EXPLORATORY PROTOTYPE · NOT CLINICAL

45%	35%	1	18
base carbs, from a PCOS meta-analysis (PMID 31885557)	floor, my assumption	scored SNP: PPARG rs1801282	tests passing

1. CYCLE PHASES (28-DAY EXAMPLE)



Ovulation is estimated as **cycle length minus 14** (a 32-day cycle gives day 18). It is a calendar estimate, not confirmed. Missing cycle data returns **insufficient_data**.

2. THE RULE

Carb % = 45 − 10 × score × (0.3 + 0.7 × luteal pressure)

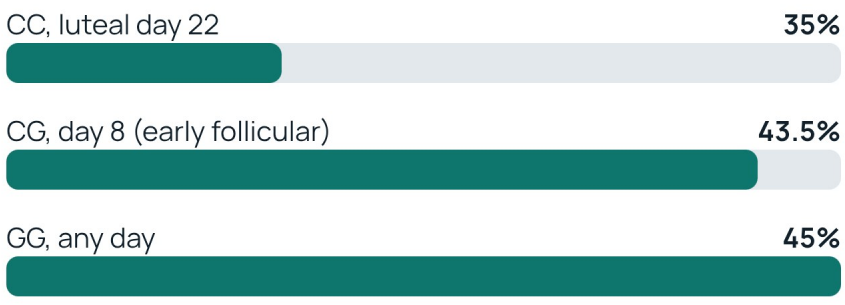
Kept between 35% and 45%.

score = weighted share of risk alleles (0 to 1)

Missing genotype = neutral, never treated as high risk

Zero-weight SNPs change nothing

WORKED EXAMPLES



Bar scale runs from 30% to 45%.

3. SNP DECISIONS

SCORED

PPARG rs1801282

Risk allele C (Pro). G (Ala) was protective in meta-analyses. Weight 1.0 is normalisation, not an effect size.

WEIGHT 0

FTO rs9939609

A is provisional. Papers disagree on direction, Indian studies were small and null, effect is BMI-mediated.

WEIGHT 0

INSR rs1799817

T risk in dominant-model, lean-women studies. Mixed: Kashmiri and South Indian studies were null.

NO ALLELE

INSR rs2059807

Papers use different strand notation, so no risk allele is assigned.

DROPPED

TPH2, ESR1

No convincing TPH2 PCOS paper. ESR1 meta-analysis was null.

CANDIDATE

KISS1 rs4889, rs372790354

Listed only. The odds ratio mapping is not confirmed.

4. WHERE EACH NUMBER COMES FROM

Sourced

45% carbs: PMID 31885557, 8 RCTs, 327 women with PCOS. A study definition, not a target.

General reference

Fiber 25 g/day and protein 0.8 g/kg from adult DRI and RDA. Not PCOS-specific.

Assumption

Genotype by phase cut, 10-point reduction, 35% floor, 0.3/0.7 mix, luteal curve, additive allele counting.

Insulin or HOMA-IR cutoff is **null**: no universal value is supported.

5. WHAT THE 18 TESTS CHECK

Day 1 maps to menstrual

28-day and 32-day cycles map correctly

Ovulation is not fixed at day 14

Missing data returns insufficient_data

High risk gives lower carbs than low risk

Carbs stay within bounds

Unknown and zero-weight SNPs change nothing

Proves the code follows its rules, not that the rules are clinically valid.

6. SAY THIS WHEN PUSHED

That part is an assumption. I labelled it. Here is how I would test it.

Validated? No. Exploratory only.

Mistake? I set FTO to T from one abstract, reversed it after comparing studies, and fixed a wrong PMID.

Not confirmed yet: PPARG PMID 26770332, plus the Gujarat and Saudi INSR PMIDs.

Amruta Rawool · github.com/amrutarawool95-spec/Cycle-Synced-Nutrition-Engine

Tools: Python, JSON, FastAPI, pytest, Colab, GitHub · Sources: dbSNP, gnomAD, Ensembl, PubMed